Feasibility Study
For PukParking

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1 Executive Summary 4
2 Problem Statement 4
2.1 Business Environment 4
2.2 Business Problem 5
3 Requirements Statement 7
3.1 Business Drivers 7
3.2 Business Requirements 7
4 Feasibility Assessment 9
4.1 Potential Solutions 9
4.2 Solution 1 - Develop a Dedicated Campus Parking Application: PukParking 9
4.3 Solution 2 - Use an Existing Parking Application: University of Pretoria (UP) 11
4.4 Solution 3 – Smart Parking Sensor System 13
5 Feasibility Ranking 16
5.1 Ranking Criteria 16
5.2 Ranking Scores 16
6 Feasibility Result 17
7 Appendix 18
7.1 Supporting Documentation 18

1 Executive Summary

The feasibility study evaluates three solutions for addressing the campus parking problem at the North-West University (NWU), Potchefstroom campus. The business problem centers around the lack of a real-time destination-based application that allows various types of users to locate available and valid parking areas, resulting in time being wasted, traffic congestion and an overall unpleasant experience for users.

The system must allow users to find a suitable and likely available parking spot close to their desired location that matches their parking disc type, through an interactive campus map and walking directions, all while displaying parking information, restrictions and allow parking spots to be marked as full.

Three solutions were assessed:

(1) Develop a dedicated campus parking application: PukParking.
(2) Use an existing parking application, such as the University of Pretoria (UP) Mobile App / existing parking functionality.
(3) Implement a Smart Parking Sensor System using IoT-based sensors to provide automated occupancy information.

Each solution was evaluated using technical feasibility, economic feasibility, operational feasibility, ability to meet user requirements, and scalability. Each criterion was scored from 1–10 and weighted according to its importance: technical feasibility (25%), economic feasibility (20%), operational feasibility (20%), user requirements (25%) and scalability (10%).

Based on the weighted feasibility scoring, Solution 1 – PukParking was chosen, scoring 8.85 out of 10. Solution 2 scored 7.40 and Solution 3 scored 6.45. PukParking achieved the highest score because it provides the strongest alignment with the identified user requirements, allows NWU-specific customisation, avoids dependence on third-party platforms and physical infrastructure, and can be deployed incrementally.

2 Problem Statement

2.1 Business Environment

The NWU Potchefstroom Campus is a large, high-density contact campus where parking demand has historically run close to capacity daily, given a campus-wide parking survey recorded occupancy of up to 92% in peak zones, and traffic modelling found volume-to-capacity ratios above 0.95 at multiple points on campus roads, indicating and contributing to existing congestion (Schoeman, 2017). Parking access is currently governed entirely manually, where students and staff purchase a physical disc, colour-coded to specific zones, and must interpret sometimes difficult

signage to understand, around campus to find a permitted available parking bay, with a R300 fine for non-compliant students, staff, and visitors (Coetzee & Du Plessis, 2022). The NWU's official guidance for locating parking areas consists of a non-interactive downloadable campus maps rather than a tool, and the university's Traffic and Access Control department, which issues discs and manages university card-based access control to campus, does not offer users any way to check real-time availability of parking or receive a destination-based recommendation of where to park their vehicle. Currently no existing tool exists, neither the NWU's own map and disc management applications nor popular general navigation apps like Google Maps or Waze offer destination-based recommendation, zone-restriction awareness, and real-time parking availability, which is the problem the PukParking project is intended to solve. Furthermore, a full breakdown of macro and micro-level environmental analysis forces is available in the Business Case document.

2.2 Business Problem

Business Problem

The Business Problem was already discussed in the Business Case document, but to summarise what was discussed:

The NWU Potchefstroom Campus faces a challenge with parking accessibility, lacking a real-time destination-based application for students, staff and visitors to locate available parking areas. Key issues discussed include high parking occupancy and the reliance on manual parking permission, which is only communicated through signage. The problem results in unnecessary vehicle movement on campus, potential fines for parking in the wrong area and new visitors having an unpleasant experience when navigating to an available parking bay. The expected date to resolve this problem is 16 October 2026.

Business Opportunity

Similar to the Business Problem, the Business Opportunity was also discussed in the Business Case document, but to summarise what was discussed:

The digitalisation of the parking allocation process at the NWU, will simplify navigation to available parking areas for students, staff and visitors by offering a real-time destination-based recommendation of available parking bays. The aim will be to reduce the time spent on searching for parking and minimise campus congestion. The initiative aligns with the NWU’s Digital Business Strategy, which supports the shift towards app-based services, while also providing administrators

with a tool to manage real-time parking data. Overall, this approach aims to enhance the overall campus experience and reduce search-time related traffic.

3 Requirements Statement

3.1 Business Drivers

The following key business drivers necessitate the completion of the PukParking solution within a defined timescale:

- Parking search efficiency: Reduce the time users spend searching for suitable and available parking on campus.
- Traffic congestion reduction: Reduce unnecessary vehicle movement caused by users driving around campus in search of parking.
- Parking disc and rule compliance: Help users identify parking areas that are valid for their parking permissions, reducing incorrect parking and potential fines.
- Improved campus accessibility and usability: Make it easier for students, staff and visitors to locate suitable parking and navigate to their destination.
- Accurate parking information: Provide users with accessible and up-to-date information about parking areas, restrictions, capacity and availability.
- Digital alignment: Support NWU's transition towards digital services by providing a digital alternative to static campus maps, physical signage and manual parking information.

3.2 Business Requirements

Business Problem or Opportunity	Project Requirement
Users have no real-time destination-based way to find a valid, available parking area.	The application that will be developed must recommend the nearest valid parking area for a selected destination, based on walking distance, parking restrictions, active status, and current occupancy.
Parking permissions are communicated only through physical parking discs and signage, not digitally.	The application must store each parking area's type and restrictions (e.g. student, staff, visitor, accessible) and only recommend areas the user is permitted to use.
There is no way to know if a recommended parking area is already full until arriving there.	Users must be able to report a parking area as full via a "Parking Full" button, which immediately recalculates and returns the next closest valid recommendation.
Repeated reports of the same full parking area risk sending many users to it in a row.	The system must maintain a temporary, crowd-sourced occupancy confidence score per parking area that lowers its

	recommendation priority and expires automatically after a configurable period.
Visitors unfamiliar with the campus have no way to navigate themselves without prior knowledge of the disc/zone system.	The application must require no registration or login and must let users search for buildings by name or abbreviation and receive turn-by-turn walking directions from the recommended parking area to the given building.
NWU's own guidance for locating parking is limited to a non-interactive PDF campus map.	The application must provide an interactive map showing all campus buildings and parking areas available to the user.
Users need to assess a parking area's suitability before navigating to it.	Each parking area must display its type, capacity, estimated occupancy, restrictions, operating hours, accessible-parking availability.
Building and parking data is currently managed manually, with no digital system of record.	The application must provide an authenticated administrator portal to add, edit, delete, and enable/disable buildings and parking areas, including capacity and restrictions.
Users need to be able to use the application on their own devices, on and off campus, without requiring it to be downloaded from any app-store.	The application must be delivered as an installable, offline-capable Progressive Web App (PWA) that is responsive across phones, tablets, and desktops.

4 Feasibility Assessment

This section documents the feasibility assessment of the three proposed alternatives for addressing the campus parking problem. The assessment considers technical, economic and operational practicality, scalability, alignment with campus requirements, implementation risks and issues. The assessment criteria are applied consistently so that the alternatives can be evaluated on their ability to support campus-specific parking discovery, building-based search, mapping, parking rules and future expansion. These requirements are particularly relevant because North-West University provides designated parking areas for students and visitors across its campuses and requires users to observe designated parking and permit rules (North-West University, 2026).

4.1 Potential Solutions

Three potential solutions were identified for addressing the campus parking problem:

- Solution 1 – Develop a Dedicated Campus Parking Application: PukParking.
- Solution 2 – Use an Existing Parking Application: University of Pretoria (UP) Mobile App / existing parking functionality.
- Solution 3 – Smart Parking Sensor System.

The alternatives differ mainly in the degree of campus-specific control, infrastructure required and implementation dependency. PukParking is assessed as a purpose-built information and navigation layer; the existing application alternative depends on the capabilities and policies of an external platform; and the sensor alternative introduces physical infrastructure to obtain automated occupancy data.

4.2 Solution 1 - Develop a Dedicated Campus Parking Application: PukParking

4.2.1 Description

PukParking is a dedicated mobile and/or web-based campus parking application designed around the needs of North-West University. It would provide students, staff and visitors with a single interface for locating suitable parking and navigating to it. The proposed core functions are:

View available or designated parking areas.

- Search for parking near a specific campus building or area.
- View parking areas on a campus map.
- Receive directions to selected parking areas.
- Identify parking zones allocated to specific user groups.
- View parking availability where reliable sensor data or controlled manual updates are available.
- Potentially reserve parking spaces where institutional policy permits.
- Receive notifications concerning parking availability, restrictions or changes.
- The underlying database would contain campus parking areas, parking spaces, buildings, users and parking rules. This structure allows the application to be

configured to NWU's actual parking layout and operating rules rather than requiring the university to adapt its processes to a generic parking product. NWU's published parking information confirms that parking is organised around designated areas, campus maps and user/permit restrictions (North-West University, 2026).

The solution can also be implemented incrementally. A first release can focus on parking discovery, campus mapping and directions, while more infrastructure-intensive features such as automated occupancy detection can be introduced later. This staged approach reduces the need for a campus-wide hardware investment at the initial implementation stage.

4.2.2 Assessment

The feasibility assessment uses five criteria: technical feasibility, economic feasibility, operational feasibility, scalability and user-requirement fit. The assessment gives particular weight to the ability to search for parking near a specific building and to represent NWU-specific parking rules. Research indicates that smart parking solutions can improve parking information, space utilisation and the user experience, while reducing inefficiencies associated with searching for available parking (Channamallu et al., 2023).

PukParking is technically feasible because the core application, database, mapping and notification components can be developed using established software technologies. Its strongest feature is the ability to define the data model and user experience around NWU's campuses. Economic feasibility is moderated by development and maintenance effort, but a phased implementation allows essential functions to be delivered before optional infrastructure is added. Operational feasibility is strengthened by a simple user-facing workflow: identify destination, view suitable parking and navigate to the selected area. Scalability is also strong because additional campuses, parking zones and functions can be added to the same platform.

Real-time occupancy should be treated as a later enhancement unless reliable sensor or other trusted data sources are available. This keeps the initial solution feasible while leaving room for integration with future smart-parking infrastructure. Research on IoT-enabled parking systems supports the feasibility of integrating sensor data with web or mobile interfaces when real-time occupancy becomes a requirement (Dheeven et al., 2024).

4.2.3 Results

Assessment Method	Feasibility Scoring
Technical Feasibility	9/10
Economic Feasibility	7/10
Operational Feasibility	9/10
Scalability	9/10
User Requirements	10/10
Overall	8.8/10

4.2.4 Risks

Risk Description	Risk Likelihood	Risk Impact	Actions Required to Mitigate Risk
Development and maintenance cost	Medium	High	Develop the solution in phases and prioritise the parking-discovery, mapping and building-search functions first.
Real-time availability	High	Medium	Begin with verified manual updates or existing trusted data sources; introduce sensor integration as a later phase.
Technical failure	Medium	High	Use reliable hosting, backups, monitoring, testing and recovery procedures.
User adoption	Medium	Medium	Provide a simple interface, clear onboarding and user guidance; use feedback to refine the application.

4.2.5 Issues

Issue Description	Issue Priority	Actions Required to Resolve Issue
Parking data must remain accurate and reflect designated areas, restrictions and changes.	High	Establish a controlled process and responsible owner for updating parking information.
The application requires an accurate and maintainable campus map.	High	Obtain approved campus maps and define a process for keeping map data current.
Future integration with access-control, sensor or other parking systems may be required.	Medium	Use modular architecture and documented APIs so future integrations can be added without redesigning the entire application.

4.2.6 Assumptions

- North-West University can provide or approve accurate information about existing parking areas and rules.
- Students, staff and visitors have access to smartphones or web-enabled devices.
- The university can provide or approve the required software infrastructure and hosting arrangements.
- The initial implementation focuses on parking discovery and navigation rather than automated physical parking control.
- Real-time parking detection can be introduced at a later stage when reliable data sources or sensors are available.

4.3 Solution 2 - Use an Existing Parking Application: University of Pretoria (UP)

4.3.1 Description

The second alternative is to use an existing parking-related application or platform rather than developing a dedicated system from the ground up. The University of Pretoria's mobile application demonstrates that an institutional mobile platform can provide campus maps, navigation-related information and parking-disc management functions (University of Pretoria, 2024). UP also operates designated parking arrangements and access controls through its security and parking services (University of Pretoria, 2026).

For NWU, this approach would require identifying an existing product that can be configured for NWU's campuses, parking areas and rules. Its principal advantage is a potentially shorter implementation period and reduced internal software-development effort. Its limitation is that NWU would depend on the selected provider or platform for the availability of the required features and the degree of customisation.

4.3.2 Assessment

The alternative is technically feasible because mature mobile and mapping functionality already exists. Economic feasibility is potentially favourable where licensing and configuration costs are lower than full in-house development. However, the assessment becomes less favourable when campus-specific requirements are considered. A generic or externally controlled application may not support NWU's exact parking zones, building-based search, user categories, data ownership requirements or future feature priorities.

This distinction is important because UP's own published information shows that its parking arrangements are closely tied to its campus rules, parking discs, access controls and designated areas (University of Pretoria, 2026). Therefore, functionality that works within UP's environment cannot automatically be assumed to transfer directly to NWU. A detailed compatibility and vendor assessment would be required before adoption.

4.3.3 Results

Assessment Method	Feasibility Scoring
Technical Feasibility	8/10
Economic Feasibility	8/10
Operational Feasibility	8/10
Scalability	7/10
User Requirements	6/10
Overall	7.4/10

4.3.4 Risks

Risk Description	Risk Likelihood	Risk Impact	Actions Required to Mitigate Risk
Limited customisation	High	High	Conduct detailed requirements analysis and product testing before adoption.
Vendor dependency	Medium	High	Review service-level agreements, support arrangements, data access and exit provisions.
Subscription or licensing costs	Medium	Medium	Compare total cost of ownership over several years before implementation.
Data limitations	Medium	High	Perform data-accuracy and coverage testing using representative NWU campus information.

4.3.5 Issues

Issue Description	Issue Priority	Actions Required to Resolve Issue
The selected application may be designed for public or commercial parking rather than NWU campus navigation.	High	Perform a detailed compatibility assessment against NWU requirements.
The application may not support building-based parking search across NWU campuses.	High	Confirm functionality through product demonstrations, testing and vendor documentation.
Parking data may remain controlled by the service provider.	Medium	Define data ownership, access, portability and retention requirements contractually.

4.3.6 Assumptions

- A suitable existing application can be identified and assessed.
- The provider permits institutional use and configuration for NWU.
- The application can support the geographical areas and campus maps required by NWU.
- The platform can satisfy most core parking-discovery requirements without extensive custom development.
- NWU accepts the level of vendor dependency and data governance associated with the selected platform.

4.4 Solution 3 – Smart Parking Sensor System

4.4.1 Description

The Smart Parking Sensor System is an IoT-based alternative that uses physical sensors to determine whether individual parking spaces are occupied. Sensor readings are transmitted to a central server and can be displayed through a mobile or web interface. An administrator dashboard can be used to monitor occupancy, manage parking areas and identify faulty sensors.

The solution would consist of parking sensors, wireless communication equipment, a backend server, database, mobile/web application and administrator dashboard. This architecture can provide highly current occupancy information and reduce dependence on manual user reports. Research confirms that sensor-enabled parking systems can transmit occupancy information to online interfaces and help users identify available spaces (Dheeven et al., 2024).

The main feasibility constraint is that the solution introduces physical infrastructure across the parking estate. The project review estimates sensor, network, software, hosting and maintenance costs and identifies campus-wide deployment as Medium-Low feasibility. A pilot in high-demand areas is therefore more practical than immediate full-campus installation.

4.4.2 Assessment

The sensor alternative was assessed using the same broad dimensions as the other solutions, with additional attention to infrastructure readiness and real-time occupancy. Smart parking research supports the value of real-time occupancy information and IoT-based monitoring, but sensor systems also introduce installation, communication, maintenance and reliability requirements (Channamallu et al., 2023; Dheeven et al., 2024).

The strongest aspect of the sensor alternative is the quality and timeliness of occupancy data once the infrastructure is operating reliably. However, the campus-wide infrastructure requirement makes initial implementation more complex. The solution therefore has greater long-term potential when integrated as a later enhancement to an established parking-information platform rather than being treated solely as the first implementation layer. A phased pilot would reduce exposure to the high initial implementation and maintenance risks identified in the project review.

4.4.3 Results

Assessment Method	Feasibility Scoring
Technical Feasibility	8/10
Economic Feasibility	7/10
Operational Feasibility	7/10
Scalability	6/10
User Requirements	8/10
Overall	7.2/10

4.4.4 Risks

Risk Description	Risk Likelihood	Risk Impact	Actions Required to Mitigate Risk
Sensors may malfunction.	Medium	High	Implement sensor monitoring, scheduled maintenance and replacement procedures.
High initial implementation cost.	High	High	Start with a limited pilot in high-demand areas before wider deployment.
Sensors may be damaged or vandalised.	Medium	High	Use durable equipment and protective installation methods.
Wireless connectivity may fail.	Medium	High	Conduct network testing and install additional gateways where required.
Sensor batteries may run out.	Medium	Medium	Schedule installation during low-traffic periods and coordinate with campus parking management.

4.4.5 Issues

Issue Description	Issue Priority	Actions Required to Resolve Issue
Installing sensors in every parking space can be expensive.	High	Begin with high-demand parking areas and expand only after pilot results justify wider deployment.
Sensors require physical maintenance.	Medium	Establish regular inspection, maintenance and replacement schedules.
Existing campus network infrastructure may not support all sensors.	High	Conduct an infrastructure and network assessment before deployment.
Sensor batteries may run out.	Medium	Schedule installation during low-traffic periods and coordinate with campus parking management.

4.4.6 Assumptions

- The university has sufficient funding for the required IoT equipment and supporting infrastructure.
- Parking spaces can be physically modified for sensor installation.
- Suitable wireless connectivity can be provided throughout the pilot and later deployment areas.
- Sensors can operate reliably under campus environmental conditions.
- Staff or contracted personnel will be available to maintain the sensor infrastructure.
- The university approves the installation of IoT equipment.
- A pilot implementation will be completed before full deployment.

4.5 Implementation Consideration

The assessment supports a staged implementation approach. The initial phase should concentrate on the functions that directly address the parking-search problem: an accurate campus parking database, building-based search, campus mapping, parking-zone information and directions. This limits initial infrastructure requirements while establishing a platform that can be tested with users.

Once the core platform is stable, additional data sources can be integrated. Real-time sensor data is a suitable later enhancement because IoT-enabled parking systems can provide automated occupancy information, but sensor deployment introduces additional hardware, network and maintenance requirements (Dheeven et al., 2024). This sequencing allows the institution to obtain useful parking-navigation functionality before committing to campus-wide sensor infrastructure.

The approach also aligns with the practical need for parking information to reflect NWU's designated areas, campus maps and parking rules (North-West University, 2026).

5 Feasibility Ranking

5.1 Ranking Criteria

Five criterias were selected to assess the feasibility of each solution.

1. **Technical feasibility:** Assess whether the solution can be realistically developed and implemented using available technology, tools and skills.
2. **Economic feasibility:** Assess whether the solution is affordable given the proposed budget and whether the benefits of the solution justify the cost.
3. **Operational feasibility:** Assesses whether the solution can be smoothly integrated into current day-to-day operations and whether the staff, students and visitors would easily adopt the solution.
4. **Ability to meet user requirements:** Assess whether the solution addresses the needs and requirements identified by the intended user**.**
5. **Scalability:** Assesses whether the solution has the ability to accommodate future growth such as additional parking without a complete redesign.

Each alternative solution was scored against every criterion on a scale of 1 - 10, where:

- 1-3: Low feasibility
- 4-6: Moderate feasibility
- 7-8: Good feasibility
- 9-10: Extremely high feasibility

Each criterion was assigned a weight to reflect its importance towards the overall project and the weights were distributed as follows:

- Technical feasibility - 25%
- Economic feasibility - 20%
- Operational feasibility - 20%
- Ability to meet user requirements - 25%
- Scalability - 10%

The maximum possible for each solution is:
(10×0.25) + (10×0.20) + (10×0.20) + (10×0.25) + (10×0.10) = **10**

5.2 Ranking Scores

	Solution 1			Solution 2			Solution 3		
Criteria	Score	Weight	Total	Score	Weight	Total	Score	Weight	Total
Technical feasibility	9	25%	2.25	8	25%	2.00	7	25%	1.75
Economic feasibility	7	20%	1.40	8	20%	1.60	4	20%	0.80
Operational feasibility	9	20%	1.80	8	20%	1.60	8	20%	1.60
Ability to meet user requirements	10	25%	2.50	9	25%	2.25	8	25%	2.00
Scalability	9	10%	0.90	7	10%	0.70	3	10%	0.30
Total Score			**8.85**			**7.40**			**6.45**
Ranking		**1st**			**3rd**			**2nd**	

6 Feasibility Result

Based on the weighted feasibility ranking, solution 1 - **Develop a Dedicated Campus Parking Application: PukParking** —achieves the highest overall score of 8.85 out of 10 and is the most feasible solution out of all solutions because it is able to satisfy the stated requirements.

Key reasons:

1. **Sustainability:** Without a third-party vendor, the system will be developed and owned by the development team and future feature additions can be handled internally without dependency on third-part platforms.
2. **Full requirements alignment:** This is the only solution able to satisfy all core requirements through consultation with the users.
3. **Reliability under peak load:** A web-based system can handle simultaneous access from hundreds of students during busy time periods better than manual reporting methods.
4. **Data-driven insights**: Built-in analytics allow the NWU campus management to identify peak congestion times and underused parking areas, supporting long-term decision making and planning.
5. **Platform independence:** A responsive web application works across all major browsers and devices without requiring an app to be downloaded, reducing both development complexity and adoption barriers for users.
6. **Swift deployment:** As no physical infrastructure needs to be procured or installed, PukParking can be developed, tested and deployed faster than hardware dependent solutions.

7 Appendix

7.1 Supporting Documentation

Attach any documentation you believe is relevant to the Feasibility Study, including:

- *Business vision, strategy and objective statements*
- *Market research documents and statistics*
- *Detailed problem analysis and requirement documentation*
- *Detailed feasibility assessment results, logs and reports*
- *Risk assessment reports*
- *Other relevant information or correspondence*

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